Marmara University
Electrical and Electronics Engineering
Probability and Statistics
Problem Sets with Solutions #5

[1] Determine the expected value of a Poisson random variable

Solution: $E[X]=\sum_{k=0}^{\infty} k\,P_X[k]=\sum_{k=0}^{\infty} k\,\frac{\lambda^k}{k!}e^{-\lambda}=e^{-\lambda}\sum_{k=0}^{\infty} k\,\frac{\lambda^k}{k!}$

$=e^{-\lambda}\left[\frac{d}{d\lambda}\left\{\sum_{k=0}^{\infty}\frac{\lambda^k}{k!}\right\}\cdot\lambda\right]=e^{-\lambda}\left[\frac{d}{d\lambda}e^{\lambda}\right]\cdot\lambda=e^{-\lambda}\,e^{\lambda}\cdot\lambda=\lambda$ ☑

[2] A discrete random variable X has the PMF $P_X[k]=\frac{1}{5}$ for $k=0,1,2,3,4$
If $Y=\sin\left[\left(\frac{\pi}{2}\right)X\right]$, find $E[Y]$ in two different ways

Solution:

1st way: $E[Y]=\sum_k k\,P_Y[k]$

X	$Y=\sin[\frac{\pi}{2}X]$
0	0
1	1
2	0
3	-1
4	0

$\Rightarrow$ $P_Y[-1]=P_X[3]=\frac{1}{5}$

$P_Y[0]=P_X[0]+P_X[2]+P_X[4]=\frac{1}{5}+\frac{1}{5}+\frac{1}{5}=\frac{3}{5}$

$P_Y[1]=P_X[1]=\frac{1}{5}$

$E[Y]=\sum_{k=-1}^{1} k\,P_Y[k]=(-1)\frac{1}{5}+0\left(\frac{3}{5}\right)+1\cdot\frac{1}{5}=0.$

2nd way: $E[Y]=\sum_k \sin\left(\frac{\pi}{2}k\right)P_X[k]$

$=\sum_{k=0}^{4}\sin\left(\frac{\pi}{2}k\right)\frac{1}{5}=\frac{1}{5}[0+1+0+(-1)+0]=0$ ☑

[3] Determine $E[X^2]$ for a geometric random variable

Solution: $E[X^2]=\sum_k k^2 P_X[k]=\sum_{k=1}^{\infty} k^2(1-p)^{k-1}\cdot p$

$\frac{d^2}{d(1-p)^2}\left\{\sum_{k=1}^{\infty}(1-p)^k\right\}=\sum_{k=1}^{\infty}k(k-1)(1-p)^{k-2}=\sum_{k=1}^{\infty}k^2(1-p)^{k-2}-\sum_{k=1}^{\infty}k(1-p)^{k-2}$

$=\frac{1}{1-p}\underbrace{\sum_{k=1}^{\infty}k^2(1-p)^{k-1}}_{\frac{E[X^2]}{p}}-\frac{1}{1-p}\underbrace{\sum_{k=1}^{\infty}k(1-p)^{k-1}}_{\frac{E[X]}{p}}$

$\frac{d^2}{d(1-p)^2}\left\{\sum_{k=1}^{\infty}(1-p)^k\right\}=\frac{1}{1-p}\frac{E[X^2]}{p}-\frac{1}{1-p}\frac{E[X]}{p}$ $\quad\left(\sum_{k=1}^{\infty}(1-p)^k=\frac{1-p}{p},\ E[X]=\frac{1}{p}\right)$

$\frac{d^2}{d(1-p)^2}\left\{\sum_{k=1}^{\infty}(1-p)^k\right\}=\frac{d^2}{dq^2}\left\{\sum_{k=1}^{\infty}q^k\right\}=\frac{d^2}{dq^2}\left\{\frac{q}{1-q}\right\}=\frac{2}{(1-q)^3}=\frac{2}{p^3}$

$1-p=q$

$\frac{2}{p^3}=\frac{E[X^2]}{p(1-p)}-\frac{1}{p^2(1-p)}\Rightarrow E[X^2]=\left[\frac{2}{p^3}+\frac{1}{p^2(1-p)}\right]p(1-p)=\frac{2-p}{p^2}=\frac{2}{p^2}-\frac{1}{p}$ ☑

[4] If $Y = aX+b$, what is the variance of Y in terms of the variance of X?

Solution: $\text{var}[Y] = \text{var}[aX+b] = E[(aX+b - E[aX+b])^2]$

$= E[(aX+b - \{aE[X]+b\})^2]$

$= E[(aX+b-aE[X]-b)^2]$

$= E[(a(X-E[X])^2]$

$= E[a^2(X-E[X])^2]$

$= a^2 E[(X-E[X])^2] = a^2 \text{var}[X]$ ☒

[5]. Find the variance of a Poisson random variable.

Solution: $\text{var}[X] = E[X^2] - (E[X])^2 = \sum_{k=0}^{\infty} k^2 P_X[k] - (\lambda)^2 = \sum_{k=0}^{\infty} k^2 \frac{\lambda^k}{k!} e^{-\lambda} - \lambda^2$

$\sum_{k=0}^{\infty} k^2 \frac{\lambda^k}{k!} e^{-\lambda} = e^{-\lambda} \sum_{k=0}^{\infty} \frac{\lambda^k}{k!}$

$\frac{d^2}{d\lambda^2}\left[\sum_{k=0}^{\infty} \frac{\lambda^k}{k!}\right] = \sum_{k=0}^{\infty} k(k-1)\frac{\lambda^{k-2}}{k!} = \sum_{k=0}^{\infty} k^2 \frac{\lambda^{k-2}}{k!} - \sum_{k=0}^{\infty} k \frac{\lambda^{k-2}}{k!}$

$= \frac{1}{\lambda^2} \underbrace{\sum_{k=0}^{\infty} k^2 \frac{\lambda^k}{k!}}_{E[X^2]\,e^{\lambda}} - \frac{1}{\lambda^2} \underbrace{\sum_{k=0}^{\infty} k \frac{\lambda^k}{k!}}_{E[X]\,e^{\lambda}} = \frac{E[X^2]e^{\lambda}}{\lambda^2} - \frac{E[X]}{\lambda^2} \cdot e^{\lambda}$

$= \frac{E[X^2]e^{\lambda}}{\lambda^2} - \frac{e^{\lambda}}{\lambda}$

$\sum_{k=0}^{\infty} \frac{\lambda^k}{k!} = e^{\lambda} \Rightarrow \frac{d^2}{d\lambda^2}\left\{\sum_{k=0}^{\infty} \frac{\lambda^k}{k!}\right\} = \frac{d^2}{d\lambda^2}\{e^{\lambda}\} = e^{\lambda}.$

$e^{\lambda} = \frac{E[X^2]e^{\lambda}}{\lambda^2} - \frac{e^{\lambda}}{\lambda} \Rightarrow E[X^2] = (1+\frac{1}{\lambda})\lambda^2 = \lambda^2+\lambda$

$\text{var}[X] = E[X^2] - \lambda^2 = \lambda^2 + \lambda - \lambda^2 = \lambda$ ☒

[6] Find the second moment ($E[X^2]$) for a Poisson random variable by using the characteristic function.

Solution: For Poisson random variable $\phi_X(\omega) = e^{\lambda(e^{j\omega}-1)}$

$E[X^2] = \frac{1}{j^2} \frac{d^2\phi_X(\omega)}{d\omega^2}\Big|_{\omega=0} = \frac{1}{j^2} \frac{d^2}{d\omega^2}\left\{e^{\lambda(e^{j\omega}-1)}\right\}\Big|_{\omega=0}$

$= -\frac{d}{d\omega}\left\{e^{\lambda(e^{j\omega}-1)} \lambda e^{j\omega}(j)\right\}\Big|_{\omega=0}$

$= -\lambda j\, e^{\lambda(e^{j\omega}-1)+j\omega}\left[\lambda e^{j\omega}(j)+j\right]\Big|_{\omega=0}$

$= -\lambda j\,[\lambda j + j] = \lambda(\lambda+1) = \lambda^2+\lambda$

☒

[7] Find the characteristic function for the PMF $P_X[k] = \frac{1}{5}$ for $k = -2, -1, 0, 1, 2$.

Solution: $\phi_x(\omega) = E[e^{j\omega x}] = \sum_{k=-2}^{2} e^{j\omega k} \cdot P_x[k] = \sum_{k=-2}^{2} e^{j\omega k} \frac{1}{5}$

$$= \frac{1}{5}\left[e^{-j2\omega} + e^{-j\omega} + 1 + e^{j\omega} + e^{j2\omega}\right]$$

$$= \frac{1}{5}\left[\frac{e^{j2\omega} + e^{-j2\omega}}{2} \cdot 2 + \frac{e^{j\omega} + e^{-j\omega}}{2} \cdot 2 + 1\right] = \frac{1}{5}\left[2\cos(2\omega) + 2\cos(\omega) + 1\right]$$

[8] Determine the variance of a binomial random variable by using the properties of the characteristic function.

Solution: $\phi_x(\omega) = (p e^{j\omega} + q)^M \quad q = 1 - p$

$$\frac{d\phi_x(\omega)}{d\omega} = M(p e^{j\omega} + q)^{M-1} p e^{j\omega}(j) \Rightarrow E[x] = \frac{1}{j}\frac{d\phi_x(\omega)}{d\omega}\Big|_{\omega=0} = Mp.$$

$$\frac{d^2\phi_x(\omega)}{d\omega^2} = M\left[(M-1)(p e^{j\omega} + q)^{M-2}(p e^{j\omega} j)^2 + (p e^{j\omega} + q)^{M-1} p e^{j\omega} j^2\right) \Rightarrow E[x^2] = \frac{1}{j^2}\frac{d^2\phi_x(\omega)}{d\omega^2}\Big|_{\omega=0} = M\left[(M-1)p^2 + p\right]$$

$$\text{Var}(x) = E[x^2] - (E[x])^2$$

$$= M^2p^2 - Mp^2 + Mp - (Mp)^2 = Mp - Mp^2 = Mp(1-p)$$

[9] Determine the mean and variance of a Poisson random variable by using the properties of the characteristic function

Solution: $\phi_x(\omega) = e^{\lambda(e^{j\omega} - 1)}$

$$\frac{d\phi_x(\omega)}{d\omega} = e^{\lambda(e^{j\omega} - 1)} \cdot \lambda e^{j\omega}(j) \Rightarrow E[x] = \frac{1}{j}\frac{d\phi_x(\omega)}{d\omega}\Big|_{\omega=0} = \frac{1}{j}(\lambda j) = \lambda$$

$$\frac{d^2\phi_x(\omega)}{d\omega^2} = e^{\lambda(e^{j\omega} - 1)}(\lambda e^{j\omega} j) + e^{\lambda(e^{j\omega} - 1)} \lambda e^{j\omega}(j)^2 \Rightarrow E[x^2] = \frac{1}{j^2}\frac{d^2\phi_x(\omega)}{d\omega^2}\Big|_{\omega=0} = \lambda^2 + \lambda$$

$$\text{Var}[x] = E[x^2] - (E[x])^2$$

$$= \lambda^2 + \lambda - (\lambda)^2$$

$$= \lambda$$